



PAPER ID-310169

Printed Page: 1 of 1
Subject Code: RCAI301

Roll No:

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MCA-INT
(SEM III) THEORY EXAMINATION 2024-25
OBJECT ORIENTED PROGRAMMING IN C++

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2*7 = 14

a.	What do you mean by virtual class?
b.	What do you mean by friend function?
c.	Write the syntax for declaring destructor?
d.	What are protected members of a class?
e.	What is operator overloading in C++?
f.	What is copy constructor?
g.	Give advantage of using inline function.

SECTION B

2. Attempt any three of the following:

7*3 = 21

a.	How Object-Oriented Programming is different from Procedural Programming?
b.	What is the use of constructors? Explain with example
c.	What is compile time polymorphism? Give example.
d.	Write a program in C++ by creating a class of prime numbers and write a function that prints all the even numbers from 1 to 100.
e.	Write a C++ program to reverse a number. Make use of member functions.

SECTION C

3. Attempt any one part of the following:

7*1 = 7

a.	Overload + operator to add two strings in C++.
b.	What is inheritance? Discuss the different types of inheritance in C++.

4. Attempt any one part of the following:

7*1 = 7

a.	Define garbage collection and compaction.
b.	What are function templates in C++? Give example.

5. Attempt any one part of the following:

7*1 = 7

a.	Differentiate between top-down and bottom-up approach.
b.	Write a program in C++ to add, multiply, divide and subtract two complex numbers using the concept of operator overloading

6. Attempt any one part of the following:

7*1 = 7

a.	What is dynamic memory allocations? Give example.
b.	Write a program to implement multilevel inheritance.

7. Attempt any one part of the following:

7*1 = 7

a.	Discuss File handling function used in C++. Explain by taking a suitable example.
b.	Write a program to handle exception in C++.